

School Database Management System



Managing student records, classes, teachers, and attendance using SQL

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PROJECT INTRODUCTION

- This project focuses on designing and managing a **School Database Management System** using **SQL**. The aim is to store, organize, and retrieve school-related data efficiently.

The project involves:

- Building a **relational database** with tables for **Students, Classes, Teachers, Subjects, and Attendance**.
- Writing **SQL queries** ranging from basic SELECT statements to advanced queries using **JOIN, GROUP BY, HAVING, and subqueries**.
- Ensuring **data integrity and relationships** using **primary keys and foreign keys**.

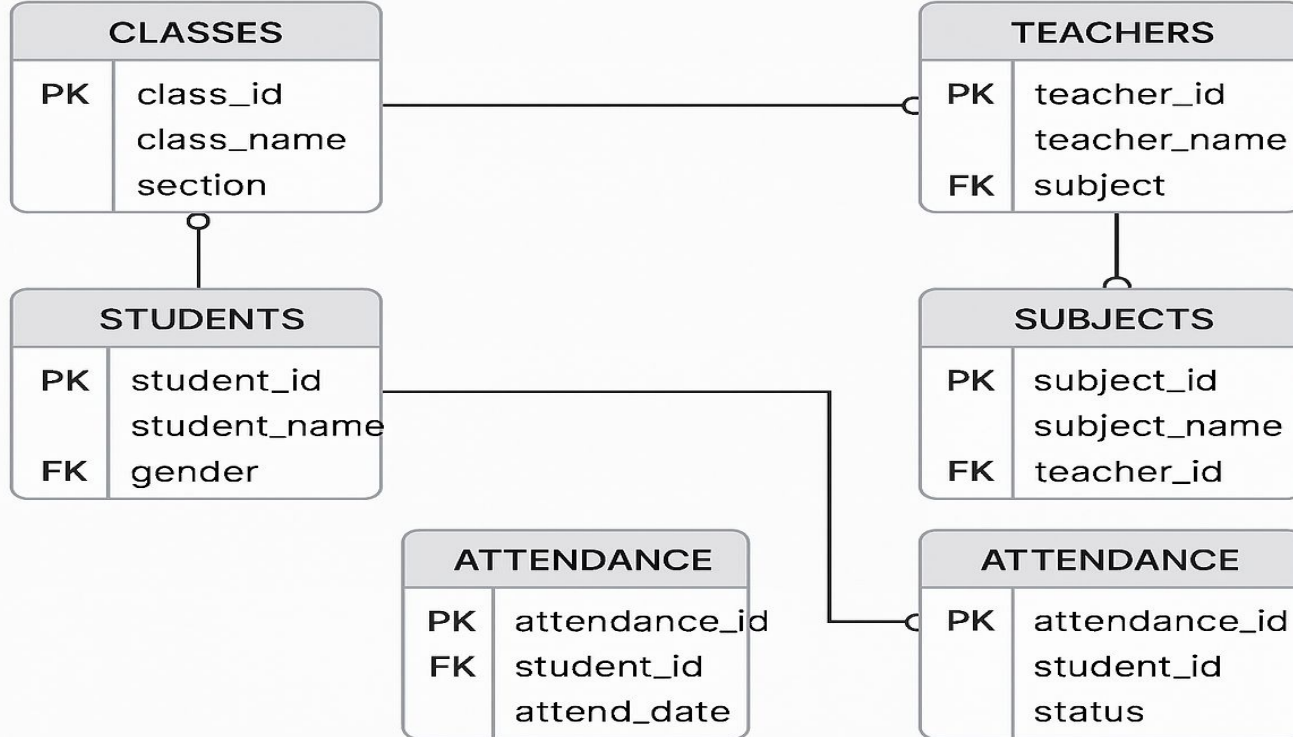
PROBLEM STATEMENT

- In many schools, managing data related to **students, classes, teachers, subjects, and attendance** is often done manually or using disconnected systems.
- This leads to problems such as **data redundancy, inconsistency, difficulty in data retrieval, and lack of proper organization**.
- The challenge is to:
- Design a **structured relational database** to store school-related data efficiently.
- Organize information about **students, classes, teachers, subjects, and attendance** in a centralized system.
- Use **SQL queries** to retrieve meaningful information such as student details, class-wise strength, teacher assignments, and attendance reports.
- Ensure **data accuracy, integrity, and easy maintenance** using primary keys and foreign keys.
- Provide a reliable database solution that supports **efficient school data management and reporting**.

PROJECT OBJECTIVE

- Build a **relational database** to manage school data efficiently.
- Design tables for **Students, Teachers, Classes, Subjects, and Attendance**.
- Implement **SQL queries** ranging from basic to advanced operations.
- Store and retrieve information related to **student records, teacher assignments, and class details**.
- Analyze data to generate **class-wise reports, student performance details, and attendance summaries**.
- Ensure **data integrity and consistency** using primary keys and foreign key relationships.

DATABASE SCHEMA (ER DIGRAM)



DATASET INFORMATION

Teachers :-

	teacher_id	teacher_name	subject
	11	Teacher 11	English
	12	Teacher 12	History
	13	Teacher 13	Maths
	14	Teacher 14	Maths
	15	Teacher 15	Geography
	16	Teacher 16	Maths
	17	Teacher 17	History
	18	Teacher 18	Maths
	19	Teacher 19	Geography
	20	Teacher 20	Maths
	21	Teacher 21	Geography
	22	Teacher 22	English
	23	Teacher 23	Maths
	24	Teacher 24	Geography
	25	Teacher 25	Science
	26	Teacher 26	Maths
	27	Teacher 27	Geography
	28	Teacher 28	Science
	29	Teacher 29	Science
	30	Teacher 30	Geography

Students :-

	student_id	student_name	gender	class_id
	257	Student 257	Female	2
	258	Student 258	Male	1
	259	Student 259	Female	8
	260	Student 260	Male	10
	261	Student 261	Female	4
	262	Student 262	Male	1
	263	Student 263	Male	7
	264	Student 264	Female	10
	265	Student 265	Male	9
	266	Student 266	Female	9
	267	Student 267	Male	10
	268	Student 268	Female	2
	269	Student 269	Female	8
	270	Student 270	Male	4
	271	Student 271	Male	7
	272	Student 272	Male	4
	273	Student 273	Male	6
	274	Student 274	Male	2
	275	Student 275	Male	10
	276	Student 276	Male	4
	277	Student 277	Female	6

Subjects :-

	subject_id	subject_name	teacher_id
▶	1	Science	26
	2	Maths	6
	3	History	5
	4	Science	4
	5	History	10
	6	History	1
	7	English	21
	8	Science	19
	9	Maths	15
	10	Geography	23
	11	Maths	4
	12	Science	26
	13	History	19
	14	English	5
	15	Geography	9
	16	Maths	23
	17	Science	8
	18	Science	11
	19	Geography	28
	20	Maths	25

Classes :-

	class_id	class_name	section
▶	1	Grade 1	A
	2	Grade 2	A
	3	Grade 3	A
	4	Grade 4	A
	5	Grade 5	A
	6	Grade 6	A
	7	Grade 7	A
	8	Grade 8	A
	9	Grade 9	A
	10	Grade 10	A

Attendance :-

	attendance_id	student_id	attend_date	status
	100	54	2025-02-09	Absent
	101	208	2025-02-28	Present
	102	12	2025-01-11	Absent
	103	141	2025-01-09	Present
	104	178	2025-02-01	Present
	105	47	2025-02-18	Present
	106	162	2025-01-11	Present
▶	107	200	2025-02-16	Absent
	108	75	2025-01-23	Present
	109	181	2025-02-21	Absent
	110	166	2025-01-18	Present
	111	168	2025-02-24	Present
	112	42	2025-02-22	Absent
	113	282	2025-02-20	Absent
	114	205	2025-01-04	Present
	115	250	2025-02-08	Absent
	116	30	2025-01-09	Absent
	117	111	2025-01-18	Absent
	118	77	2025-01-16	Present
	119	208	2025-01-01	Present
	120	28	2025-01-29	Present
	121	274	2025-01-10	Absent

QUERIES CLASSIFICATION

- Simple Queries:

Basic SELECT, WHERE, ORDER BY

- Intermediate Queries:

GROUP BY, HAVING, Aggregates

- Advanced Queries:

Joins, Aggregations

Simple Queries

1. Student name and Gender

```
9      -- 1.Display the student_name, and gender of all students.
10 •   select student_name,gender from students;
11
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	student_name	gender			
▶	Student 1	Male			
	Student 2	Male			
	Student 3	Male			
	Student 4	Male			
	Student 5	Male			
	Student 6	Female			
	Student 7	Female			
	Student 8	Male			
	Student 9	Female			
	Student 10	Female			
	Student 11	Female			
	Student 12	Female			
	Student 13	Female			
	Student 14	Male			
	Student 15	Male			
	Student 16	Male			
	Student 17	Female			

2. Classes ordered by class name

```
12      -- 2.List all classes ordered by class_name.  
13 •    select * from classes order by class_name;
```

Result Grid




Filter Rows:

Edit:





	class_id	class_name	section
▶	1	Grade 1	A
	10	Grade 10	A
	2	Grade 2	A
	3	Grade 3	A
	4	Grade 4	A
	5	Grade 5	A
	6	Grade 6	A
	7	Grade 7	A
	8	Grade 8	A
	9	Grade 9	A
✱	NULL	NULL	NULL

3. Teachers name for maths subject

```
15 -- 3.Display the names of teachers who teach a Maths.  
16 • select teacher_name,subject from teachers where subject = 'Maths';
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	teacher_name	subject			
▶	Teacher 1	Maths			
	Teacher 7	Maths			
	Teacher 13	Maths			
	Teacher 14	Maths			
	Teacher 16	Maths			
	Teacher 18	Maths			
	Teacher 20	Maths			
	Teacher 23	Maths			
	Teacher 26	Maths			

4. Students with class_id = 7

```
19      -- 4.Show all students who belong to a class_id is 7.  
20 •    select * from students where class_id = 7;
```

Result Grid |   Filter Rows: | Edit:    | Export/In

	student_id	student_name	gender	class_id
▶	2	Student 2	Male	7
	4	Student 4	Male	7
	14	Student 14	Male	7
	27	Student 27	Female	7
	34	Student 34	Female	7
	36	Student 36	Male	7
	37	Student 37	Male	7
	39	Student 39	Female	7
	48	Student 48	Male	7
	54	Student 54	Female	7
	65	Student 65	Male	7
	74	Student 74	Female	7
	83	Student 83	Female	7
	84	Student 84	Female	7
	86	Student 86	Female	7
	94	Student 94	Male	7

5. Attendance records for after january

```
22      -- 5.Retrieve all attendance records for a after january.  
23 •    select * from attendance where attend_date > '2025-01-31';
```

Result Grid   Filter Rows: Edit:    Export/Import:				
	attendance_id	student_id	attend_date	status
▶	2	292	2025-02-24	Absent
	5	208	2025-02-10	Present
	7	17	2025-02-03	Absent
	12	290	2025-02-20	Absent
	13	70	2025-02-14	Present
	18	28	2025-02-05	Present
	19	156	2025-02-04	Absent
	21	160	2025-02-24	Present
	22	70	2025-02-07	Absent
	23	221	2025-02-21	Present
	24	21	2025-02-25	Present
	25	124	2025-02-16	Absent
	26	300	2025-02-27	Present
	27	230	2025-02-06	Absent

Intermediate Queries

1. Students count for each class

```
27 -- 1.Count the total number of students in each class.  
28 • select class_id,count(student_name) from students group by class_id;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	class_id	count(student_name)		
▶	1	36		
	2	26		
	3	30		
	4	17		
	5	31		
	6	40		
	7	36		
	8	27		
	9	25		
	10	32		

2. Number of students by gender

```
30      -- 2.Display the number of students by gender.  
31 •    select gender,count(student_name) from students group by gender;
```

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
	gender	count(student_name)			
▶	Male	137			
	Female	163			

3. Find the subjects for more than 5 teachers

```
33      -- 3.Find the subjects that are taught by more than 5 teacher.  
34 •    select subject,count(teacher_id) from teachers group by subject having count(teacher_id) > 5;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	subject	count(teacher_id)			
▶	Maths	9			
	Science	6			
	Geography	7			

4. Number of students present on each date

```
36 -- 4.Find the number of students present on each date.
37 • select attend_date,count(attendance_id) from attendance where status = 'Present' group by attend_date;
```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	attend_date	count(attendance_id)
▶	2025-01-15	1
	2025-01-05	1
	2025-02-10	3
	2025-01-19	1
	2025-02-14	2
	2025-01-24	1
	2025-02-05	3
	2025-01-02	2
	2025-02-24	3
	2025-02-21	3
	2025-02-25	2
	2025-02-27	2
	2025-01-12	4
	2025-02-20	1

5. Class_id for more than 10 students

```
39 -- 5.Display class IDs having more than 10 students.  
40 • select class_id,count(student_id) from students group by class_id having count(student_id) > 10;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	class_id	count(student_id)
▶	1	36
	2	26
	3	30
	4	17
	5	31
	6	40
	7	36
	8	27
	9	25
	10	32

Advanced Queries

1. Student_name and Class_name and

```
44 -- 1.Display student names along with their class name and section.  
45 • select s.student_name,c.class_name,c.section from students s left join classes c  
46 on s.class_id = c.class_id;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
student_name	class_name	section	
Student 1	Grade 1	A	
Student 2	Grade 7	A	
Student 3	Grade 6	A	
Student 4	Grade 7	A	
Student 5	Grade 9	A	
Student 6	Grade 10	A	
Student 7	Grade 8	A	
Student 8	Grade 5	A	
Student 9	Grade 8	A	
Student 10	Grade 6	A	
Student 11	Grade 6	A	
Student 12	Grade 1	A	
Student 13	Grade 5	A	
Student 14	Grade 7	A	

2. Subject name and teacher name

```
48 -- 2.List subjects along with the teacher name who teaches them, sorted by teacher name in ascending order.
49 • select s.subject_name,t.teacher_name from subjects s left join teachers t
50 on s.teacher_id = t.teacher_id order by teacher_name asc;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
subject_name	teacher_name		
History	Teacher 1		
History	Teacher 10		
Science	Teacher 11		
Maths	Teacher 15		
Science	Teacher 19		
History	Teacher 19		
English	Teacher 21		
Geography	Teacher 23		
Maths	Teacher 23		
Maths	Teacher 25		
Science	Teacher 26		
Science	Teacher 26		
Geography	Teacher 28		
Science	Teacher 4		

3. Subject name with attendance status for each date

```
52 -- 3.Display student names and their attendance status for each date, sorted by attendance date in ascending order.
53 • select s.student_name,a.attend_date,a.status from students s right join attendance a
54 on a.student_id = s.student_id order by attend_date asc;
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	student_name	attend_date	status
▶	Student 208	2025-01-01	Present
	Student 71	2025-01-01	Absent
	Student 5	2025-01-01	Present
	Student 290	2025-01-01	Present
	Student 167	2025-01-02	Present
	Student 159	2025-01-02	Present
	Student 182	2025-01-02	Absent
	Student 93	2025-01-03	Absent
	Student 164	2025-01-03	Absent
	Student 69	2025-01-04	Absent
	Student 54	2025-01-04	Absent
	Student 63	2025-01-04	Present
	Student 205	2025-01-04	Present
	Student 234	2025-01-04	Absent

4. Class name has the highest number of students

```
56 -- 4.Find the class name that has the highest number of students.  
57 • with CTE_1 as(select class_id,count(student_id) from students group by class_id order by count(student_id) desc limit 1)  
58 select c.class_name from CTE_1 s left join classes c  
59 on s.class_id = c.class_id;
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 

	class_name
▶	Grade 6

5. Absent Students list

```
61  -- 5.List students who were marked Absent at least once.
62  •  with cte_1 as(select * from students),
63      cte_2 as(select student_id,status from attendance where status = 'Absent')
64      select s.*,a.status from cte_2 a right join cte_1 s
65      on a.student_id = s.student_id where a.status = 'Absent';
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	student_id	student_name	gender	class_id	status
	54	Student 54	Female	7	Absent
	238	Student 238	Male	7	Absent
	153	Student 153	Female	6	Absent
	219	Student 219	Male	2	Absent
	87	Student 87	Female	6	Absent
	247	Student 247	Male	10	Absent
	29	Student 29	Male	1	Absent
	86	Student 86	Female	7	Absent
	279	Student 279	Female	5	Absent
	88	Student 88	Female	6	Absent
	242	Student 242	Male	5	Absent
	182	Student 182	Female	9	Absent
	138	Student 138	Male	1	Absent
	164	Student 164	Female	10	Absent

KEY INSIGHTS

- **Student strength differs across classes**, with some classes having significantly more students than others.
- **Gender distribution is fairly balanced**, but varies slightly by class.
- **Maths and other core subjects are taught by multiple teachers**, showing high academic demand.
- **Attendance is mostly consistent**, though **several students were marked absent at least once**.
- **A few classes exceed 10 students**, indicating higher classroom density and resource needs.

PROJECT BENEFITS

- **Strengthened SQL skills** using real-world school data
- **Practical experience with JOINS, GROUP BY & CTEs**
- **Clear insights into students, classes & attendance**
- **Improved data analysis and problem-solving ability**
- **Project suitable for academic submission & resume**

CONCLUSION

- **SQL helped transform raw school data into meaningful insights**, improving data-driven understanding.
- **Student, class, teacher, and attendance data were effectively analyzed** using SQL queries.
- **JOINS, GROUP BY, HAVING, and CTEs enabled deeper analysis** across multiple tables.
- **Attendance patterns and class strength trends were clearly identified**, supporting better academic monitoring.
- **This project proves SQL as a powerful tool for analyzing real-world educational datasets.**

Thank you